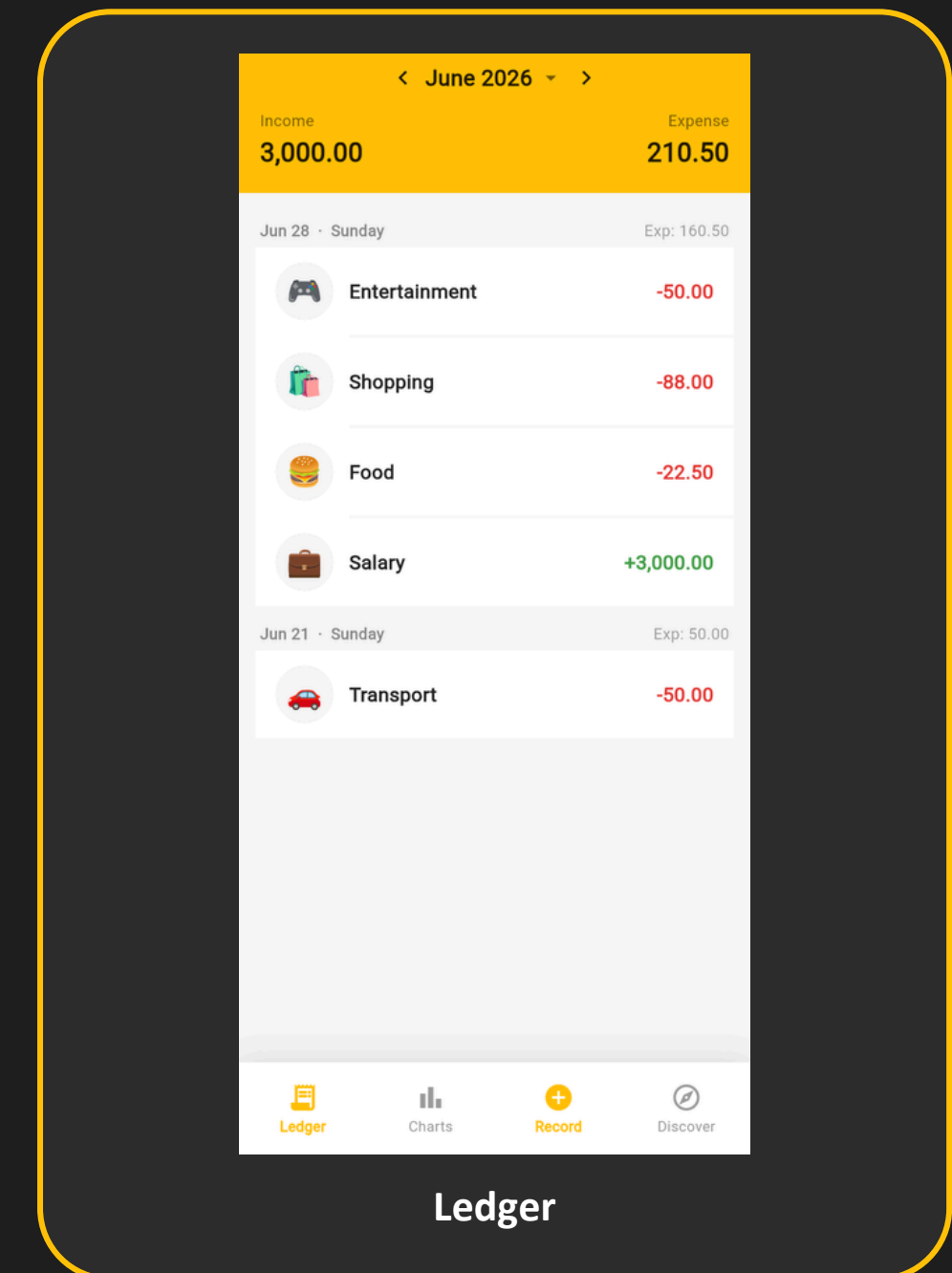


My Wallet

Personal Expense Tracker

A cross-platform CRUD mobile app built with Flutter & Firebase

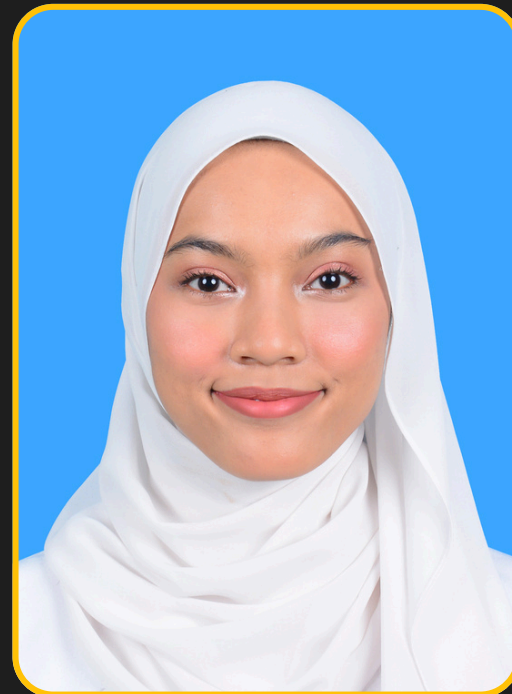
DES3113 — Mobile App Design & Development · Project Pitch
Team of 5 · Lecturer: Dr. Ahmad Wiraputra bin Selamat · 2026



Group Members



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■ The Problem

Where does my money actually go?

Small, frequent purchases are easy to forget and quickly add up. Most finance apps are cluttered, ad-heavy, or hide the basics behind a paywall.



Hard to track

Daily cash spending slips through the cracks.



Too complex

Existing apps are bloated and confusing.



No backup

Records trapped on one phone — lost if it's gone.



No insight

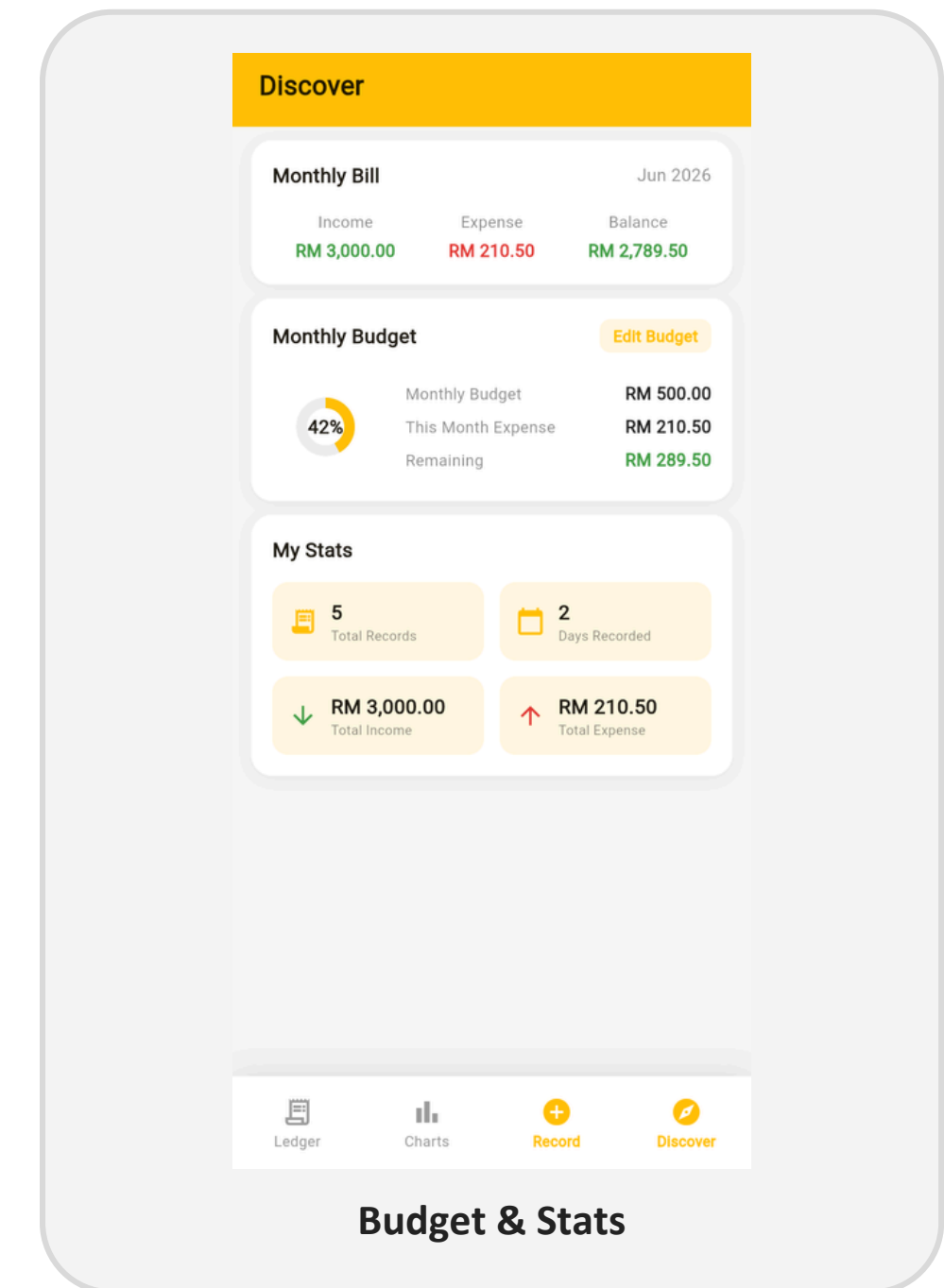
Spending patterns and budgets aren't obvious.

■ The Solution

A fast, simple, cloud-backed ledger

My Wallet lets you record income and expenses in seconds, browse them by day and month, see spending trends, and stay on budget — with every record stored safely in the cloud and synced in realtime.

- ✓ Record a transaction in just a few taps
- ✓ Browse by day, switch between months
- ✓ Charts and category breakdown
- ✓ Monthly budget tracking
- ✓ Realtime cloud sync with Firebase



■ Key Features



Smart Ledger

Transactions grouped by day with monthly totals.



Quick Record

Custom keypad, category grid, date & note.



Charts

Week / Month / Year trend + category breakdown.



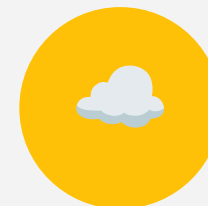
Budget

Set a monthly limit; track remaining balance.



Statistics

Lifetime income, expense, balance & activity.



Cloud Database

Stored in Cloud Firestore, synced in realtime.

■ CRUD at the Core

Every requirement maps to a real feature, all driven by one Provider.



Create

Add a record via the keypad,
category grid, date picker and
note.



Read

Browse the Ledger grouped by
day; open any record in Detail.



Update

Edit a record — the Add screen
reopens pre-filled.



Delete

Remove a record with a confirm
dialog, or clear all.

■ Architecture

A clean, layered design — the UI never touches storage directly.

UI Layer Ledger · Charts · Record · Discover · Add / Edit

↓ *context.watch / read*

State Layer TransactionProvider (ChangeNotifier) — CRUD, totals, budget

↓ *toMap / fromMap*

Model Layer Transaction · Category · TransactionType

↓ *snapshots / set / delete*

Storage Layer Cloud Firestore (realtime NoSQL database)

Why it works

- Single source of truth
- UI rebuilds on change
- Logic easy to test
- Realtime cloud sync

Technology Stack

Flutter

Cross-platform UI toolkit (Material 3)

Dart

Application language (SDK ^3.11)

firebase_core

Firebase initialisation

cloud_firestore

Realtime cloud database (CRUD)

provider

ChangeNotifier state management

intl · uuid

Formatting & unique IDs

flutter_test

Unit & widget testing

Git + GitHub

Version control & collaboration

■ State & Persistence

Provider as the single source of truth

TransactionProvider extends ChangeNotifier. Each CRUD method mutates the list, saves to storage, then calls notifyListeners() so every watching widget rebuilds automatically.

Cloud persistence

Each transaction is a document in Cloud Firestore. A realtime snapshot listener streams changes back, so the UI updates instantly across devices.

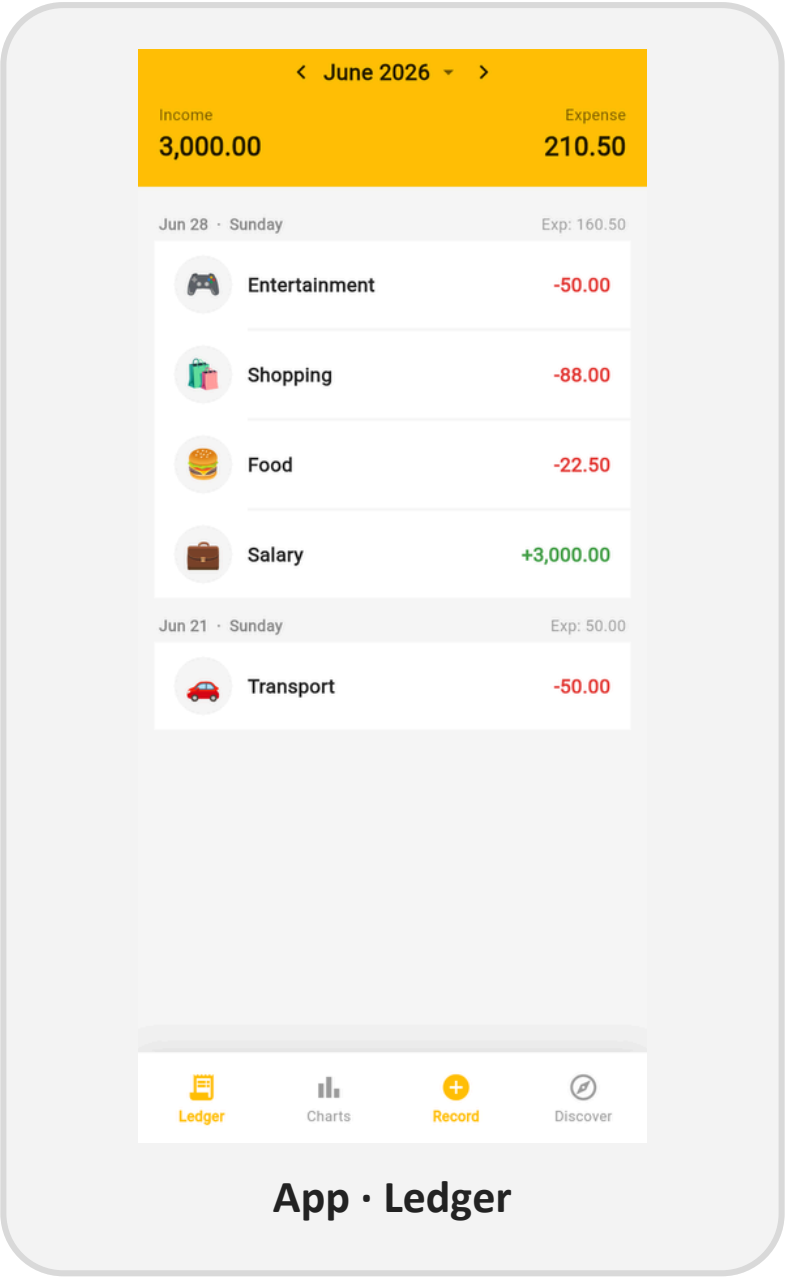
```
// Realtime read
_txCol.snapshots().listen((snap) {
  _transactions = snap.docs
    .map((d) =>
      Transaction.fromMap(d.data()))
    .toList();
  notifyListeners();
});

// Create / Update
await _txCol.doc(t.id).set(t.toMap());

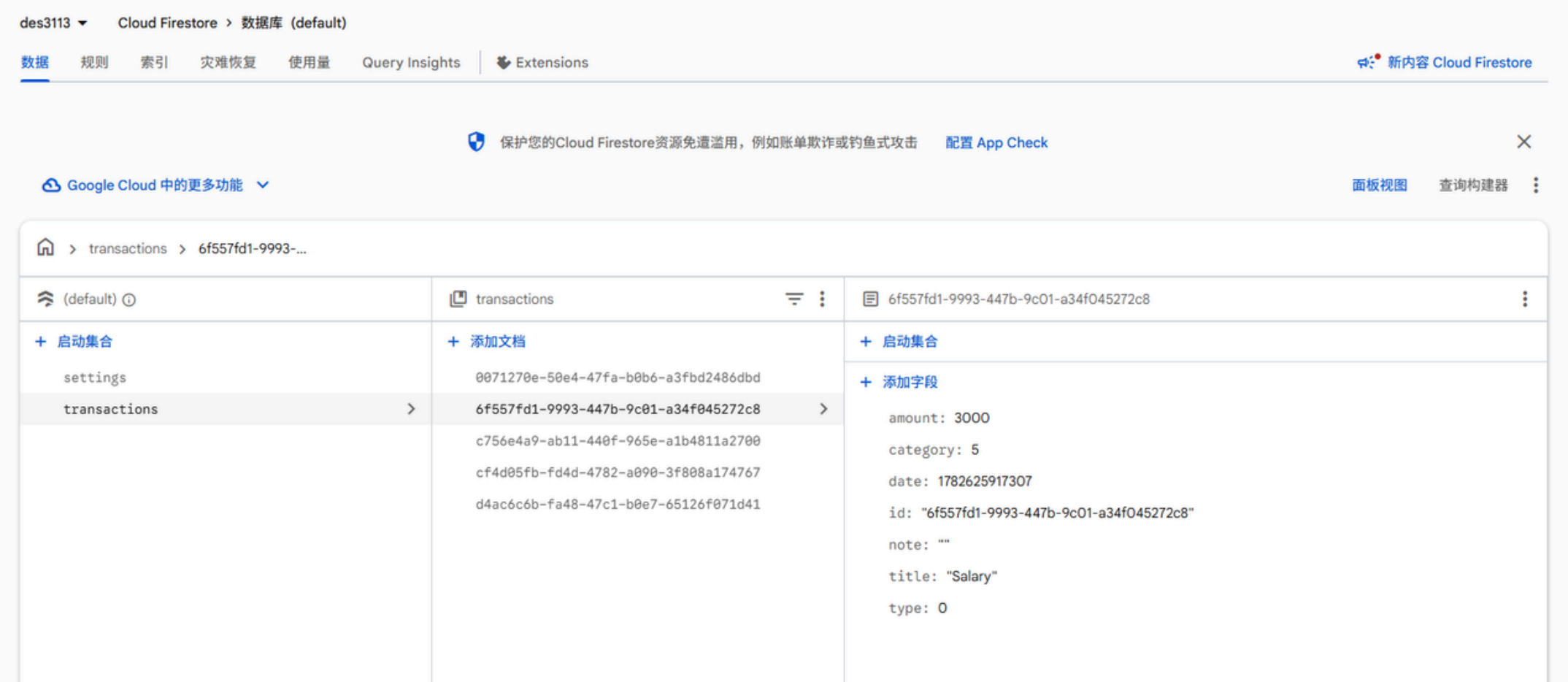
// Delete
await _txCol.doc(id).delete();
```

Cloud Database — Firebase

Every record is a document in Cloud Firestore — created, updated and deleted in realtime.



Realtime sync

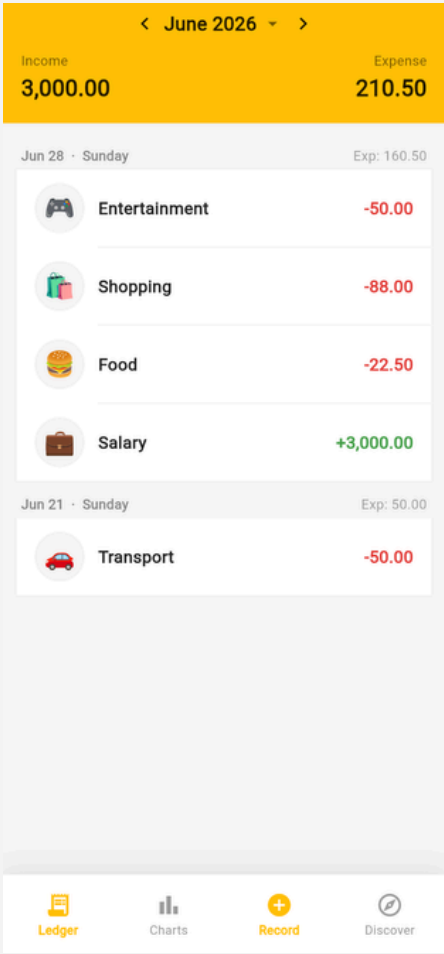


Cloud Firestore · project des3113-7d1fe · transactions collection

transactions — one document per record; Create/Update via `set()`, Delete via `delete()`.
Fields — amount · category · type · date · title · note, mirroring the model.

■ UI / UX Design

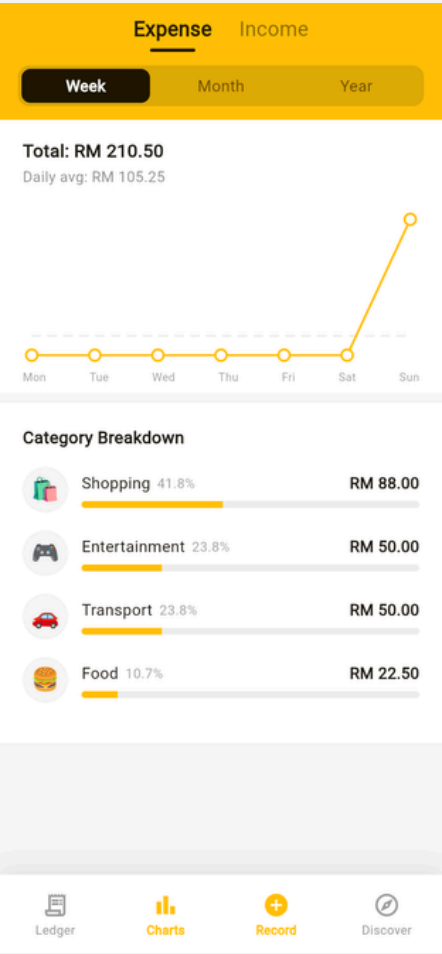
Material 3 · amber brand colour · familiar daily-ledger layout.



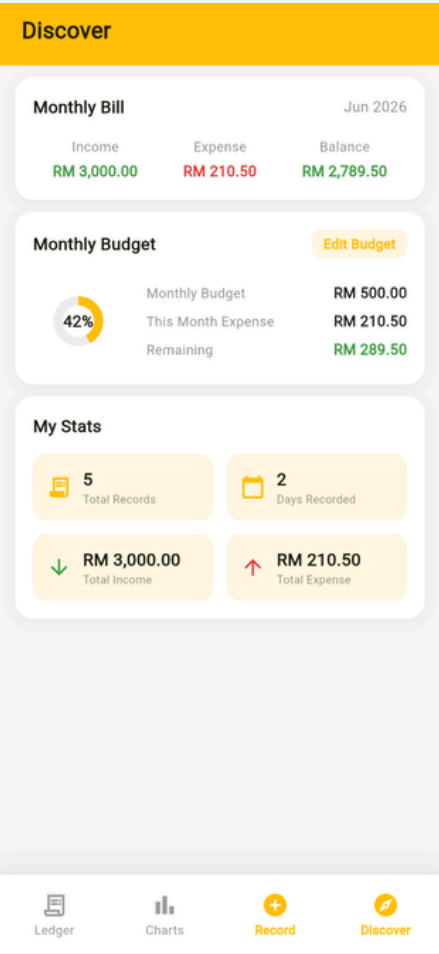
Ledger

The Add Record screen is for entering a new expense. It features a top bar with 'Expense', 'Income', and 'Cancel' options. The current amount is 'RM 45.80'. Below this is a date selector for 'Jun 28, 2026'. A category selection grid includes Food, Transport, Shopping, Entertainment, Health, Education, and Other. There's a text input field for 'Add a note...' and a numeric keypad at the bottom with a 'Save' button. The bottom navigation bar is the same as the Ledger screen.

Add Record



Charts



Discover

■ Testing

12 / 12

automated tests passing

What we tested



Unit — Model

Serialisation round-trips, copyWith, category labels.



Unit — Provider

CRUD + budget on an in-memory test database.



Widget

App builds and the bottom navigation renders.



Manual

9 flows incl. data appearing live in the Firebase console.

■ Team & Contributions

1

Sazit UI Islam

Project lead · State management & data model

2

Nur Aisyatul Najwa binti Mohamad Nasir

UI/UX · Ledger & Detail screens

3

Nurfatin Aqilah binti Zolkifli

Add/Edit & Charts screens

4

Shi Kaiyan

Discover screen · Firebase & testing

5

Mst Samiya Haque Kotha

Documentation, report & presentation

Every member contributed across design, development, testing & docs. Source managed with Git + GitHub.

■ Challenges & What's Next

Challenges we solved

- ✓ Keeping UI in sync → single Provider
- ✓ Cloud data → Firestore + realtime listener
- ✓ Trend chart → custom CustomPainter
- ✓ Valid input → keypad guard rules

Future enhancements

- User accounts (Firebase Auth)
- Firestore security rules
- Export to CSV / PDF
- Recurring bills & reminders
- Dark mode & multi-currency

Thank You

My Wallet — Personal Expense Tracker

Built with Flutter · Firebase · Cloud Firestore